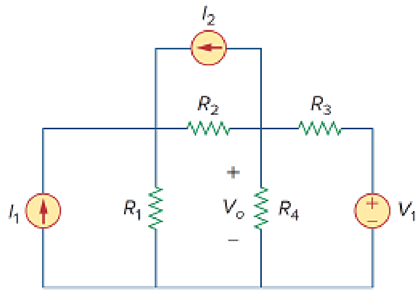
Chapter 3, Problem 68P

Problem

Using Fig. 3.112, design a problem, to solve for *Vo*, to help other students better understand nodal analysis. Try your best to come up with values to make the calculations easier.

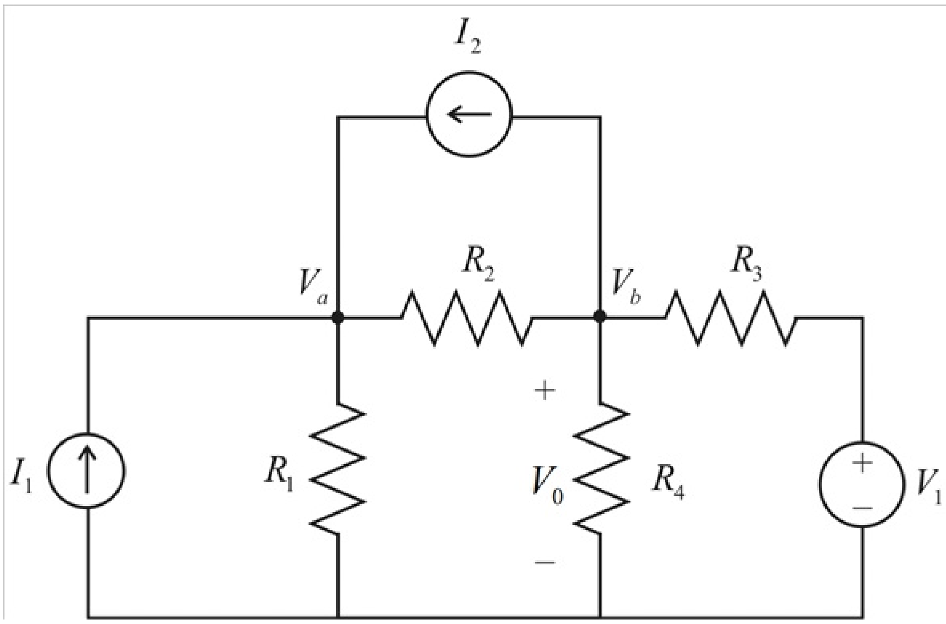
Figure 3.112:



Step-by-step solution

Step 1 of 7

Considering the circuit



Step 2 of 7

By applying KCL at node a we get

$$-I_1+\frac{V_a}{R_1}+\frac{V_a-V_b}{R_2}-I_2=0$$

$$-I_1+\frac{V_a}{R_1}+\frac{V_a}{R_2}+\frac{-V_b}{R_2}-I_2=0$$

$$-I_1+V_a\left(\frac{1}{R_1}+\frac{1}{R_2}\right)-\frac{V_b}{R_2}-I_2=0$$

$$-I_1+V_a\left(\frac{R_1+R_2}{R_1R_2}\right)-\frac{V_b}{R_2}-I_2=0 \quad \ldots\ldots \text{(1)}$$

By applying KCL at node b we get

$$I_2+\frac{V_b}{R_4}+\frac{V_b-V_a}{R_2}+\frac{V_b-V_1}{R_3}=0$$

$$I_2+\frac{V_b}{R_4}+\frac{V_b}{R_2}+\frac{V_b}{R_3}-\frac{V_a}{R_2}-\frac{V_1}{R_3}=0$$

$$I_2+V_b\left(\frac{1}{R_4}+\frac{1}{R_2}+\frac{1}{R_3}\right)-\frac{V_a}{R_2}-\frac{V_1}{R_3}=0 \text{ ...... (2)}$$

By simplifying equation (1) we get

$$-R_1R_2I_1+V_aR_2+R_1V_a-R_1V_b-R_1R_2I_2=0$$

$$V_a\left(R_1+R_2\right)=R_1R_2I_1+R_1V_b+R_1R_2I_2$$

$$V_a=\frac{R_1R_2I_1+R_1V_b+R_1R_2I_2}{R_1+R_2} \text{ ...... (3)}$$

Substituting V_a in equation (2) then we get

$$I_2+V_b\left(\frac{1}{R_4}+\frac{1}{R_2}+\frac{1}{R_3}\right)-\frac{V_a}{R_2}-\frac{V_1}{R_3}=0$$

$$I_2+V_b\left(\frac{R_2R_3+R_4R_3+R_2R_4}{R_2R_3R_4}\right)-\frac{V_a}{R_2}-\frac{V_1}{R_3}=0$$

$$I_2R_2R_3R_4+V_b\left(R_2R_3+R_4R_3+R_2R_4\right)-V_aR_3R_4-V_1R_2R_4=0$$

From (3) we have $V_a=\dfrac{R_1R_2I_1+R_1V_b+R_1R_2I_2}{R_1+R_2}$

By substituting this value we get

$$I_2R_2R_3R_4+V_b\left(R_2R_3+R_4R_3+R_2R_4\right)-\left(\frac{R_1R_2I_1+R_1V_b+R_1R_2I_2}{R_1+R_2}\right)R_3R_4-V_1R_2R_4=0$$

$$\Rightarrow R_2R_3R_4I_2+R_2R_3V_b+R_3R_4V_b+R_2R_4V_b-R_3R_4\left[\frac{R_1R_2I_1+R_1V_b+R_1R_2I_2}{R_1+R_2}\right]-R_2R_4V_1=0$$

$$\Rightarrow R_2R_3R_4I_2+R_2R_3V_b+R_3R_4V_b+R_2R_4V_b+\frac{-R_3R_4R_1R_2I_1-R_3R_4R_1R_2V_b-R_3R_4R_1R_2I_2}{R_1+R_2}$$
$$-R_2R_4V_1=0$$

$$\Rightarrow (R_1+R_2)(R_2R_3R_4I_2)+(R_1+R_2)(R_2R_3V_b)+(R_1+R_2)(R_3R_4V_b)+(R_1+R_2)(R_2R_4V_b)$$
$$-R_3R_4R_1R_2I_1-R_3R_4R_1V_b-R_3R_4R_1R_2I_2-(R_1+R_2)(R_2R_4V_1)=0$$

$$\Rightarrow R_1R_2R_3R_4I_2+R_2^{\,2}R_3R_4I_2+R_1R_2R_3V_b+R_2^2R_3V_b+R_1R_3R_4V_b+R_2R_3R_4V_b$$
$$+R_1R_2R_4V_1+R_2^2R_4V_1-R_1R_2R_3R_4I_1-R_1R_3R_4V_b-R_1R_2R_3R_4I_2+R_1R_2R_4V_b$$
$$+R_2^2R_4V_b+R_1R_2R_3I_2+R_2R_3R_4I_2=0$$

$$\Rightarrow R_2^{\,2}R_3R_4I_2+R_1R_2R_3V_b+R_2^2R_3V_b+R_2R_3R_4V_b+R_1R_2R_4V_1+R_2^2R_4V_1$$
$$-R_1R_2R_3R_4I_1+R_1R_2R_4V_b+R_2^2R_4V_b+R_1R_2R_3I_2+R_2R_3R_4I_2=0$$

$$\Rightarrow R_2^{\,2}R_3R_4I_2+\left(R_1R_2R_3+R_2^2R_3+R_2R_3R_4+R_1R_2R_4+R_2^2R_4\right)V_b-R_1R_2R_4V_1-R_2^2R_4V_1$$
$$-R_2^{\,2}R_3R_4I_2-R_1R_2R_3R_4I_1=0$$

$$\Rightarrow \left(R_1R_2R_3+R_2^2R_3+R_2R_3R_4+R_1R_2R_4+R_2^2R_4\right)V_b=\begin{pmatrix}-R_2^{\,2}R_3R_4I_2+R_1R_2R_4V_1+\\ R_2^2R_4V_1+R_1R_2R_3R_4I_1\end{pmatrix}$$

$$\Rightarrow V_b=\frac{-R_2^{\,2}R_3R_4I_2+R_1R_2R_4V_1+R_2^2R_4V_1+R_1R_2R_3R_4I_1}{\left(R_1R_2R_3+R_2^2R_3+R_2R_3R_4+R_1R_2R_4+R_2^2R_4\right)}$$

From the circuit the output voltage is

$$\Rightarrow V_0=V_b$$

$$\therefore \boxed{V_0=\frac{-R_2^{\,2}R_3R_4I_2+R_1R_2R_4V_1+R_2^2R_4V_1+R_1R_2R_3R_4I_1}{\left(R_1R_2R_3+R_2^2R_3+R_2R_3R_4+R_1R_2R_4+R_2^2R_4\right)}}$$